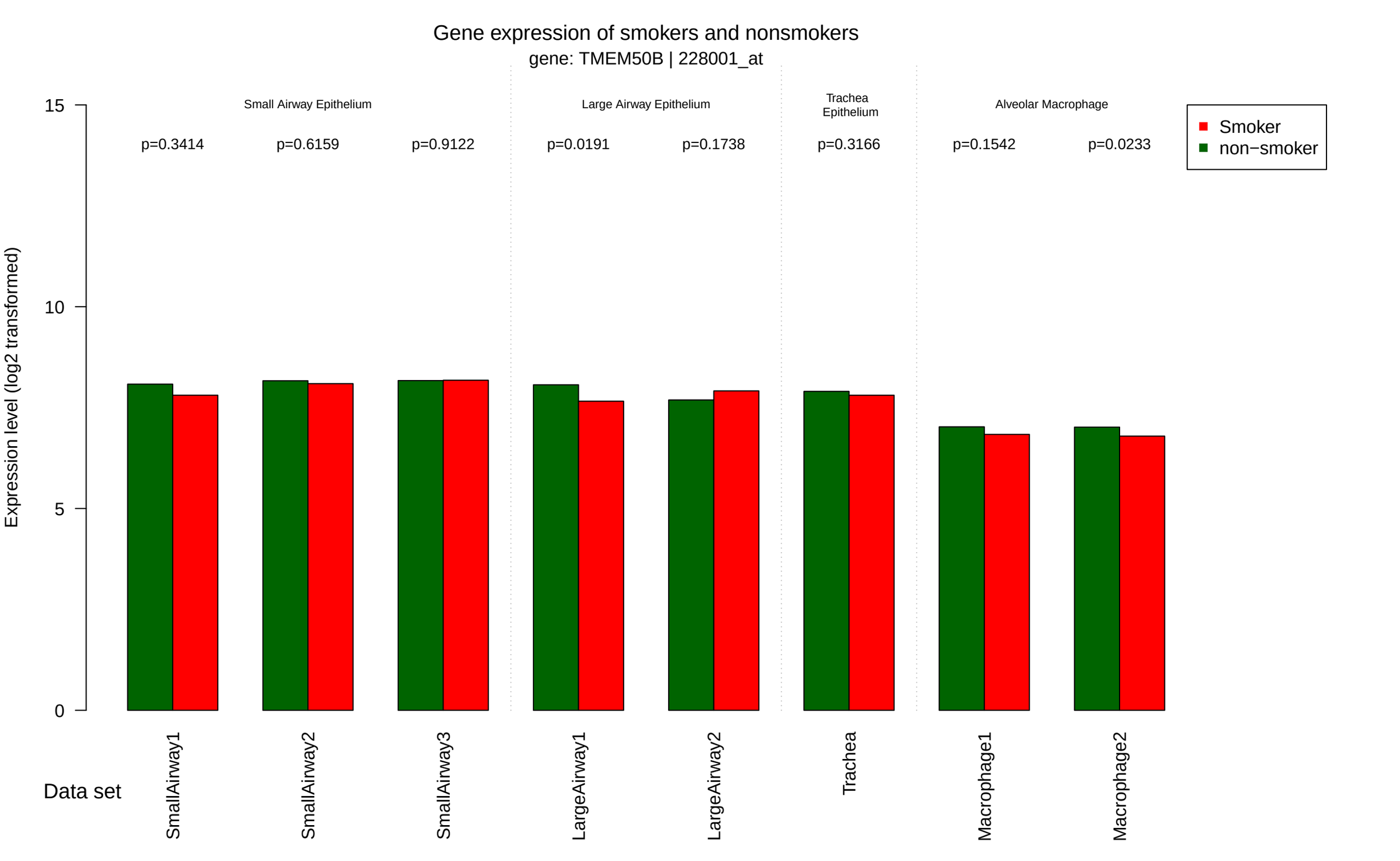
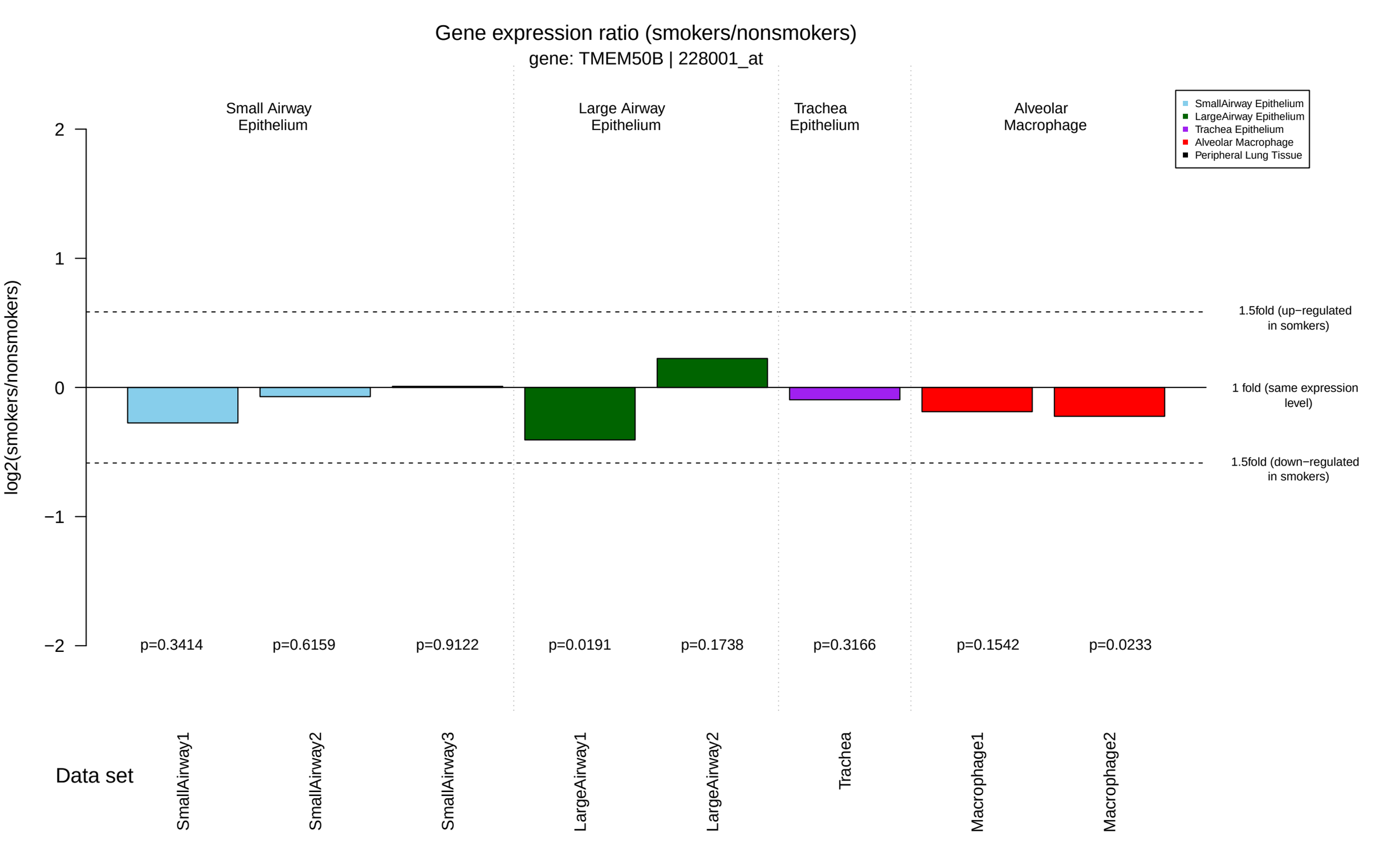
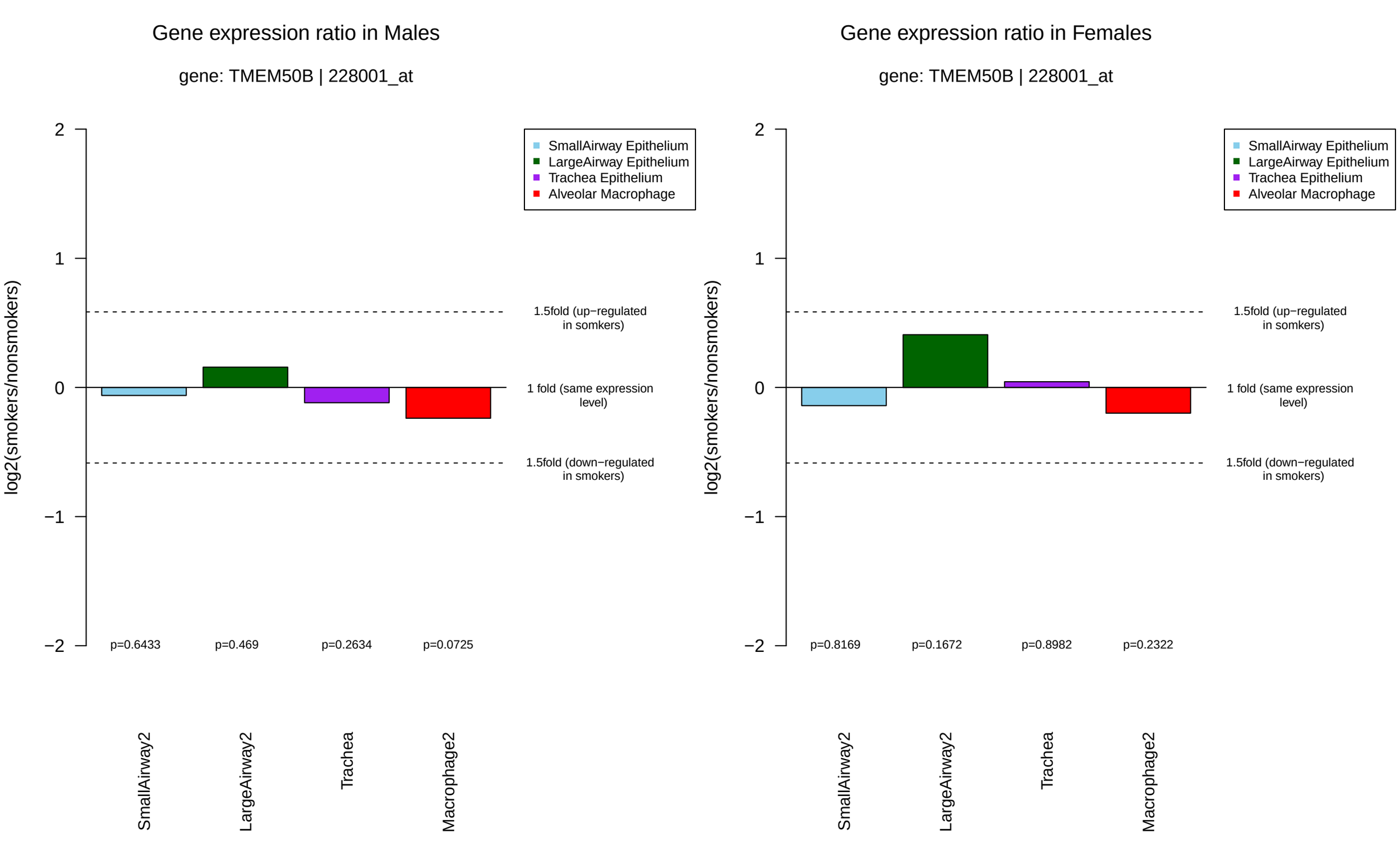
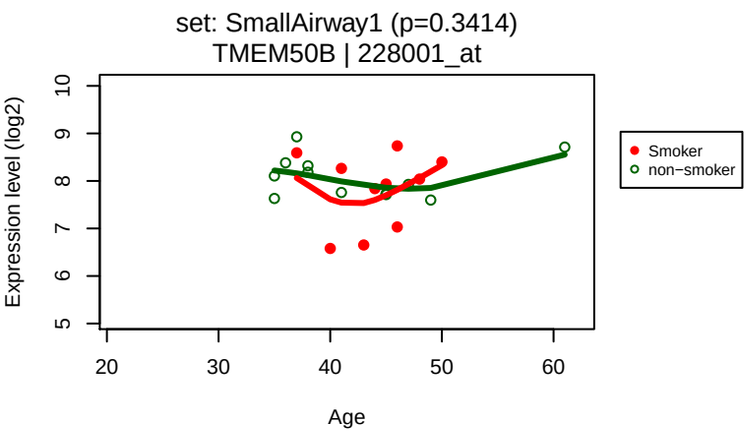
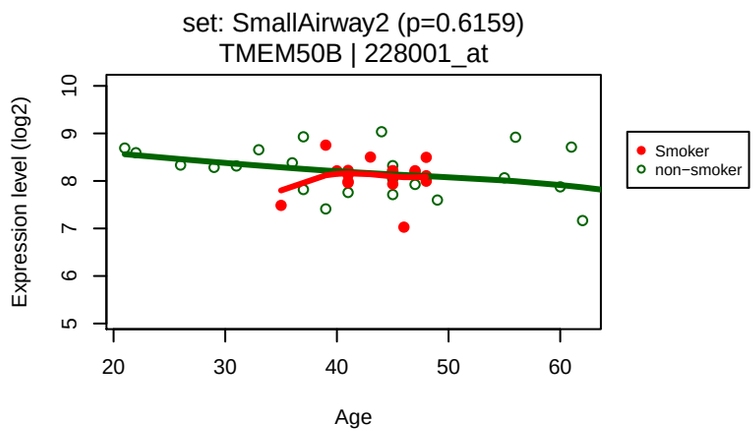
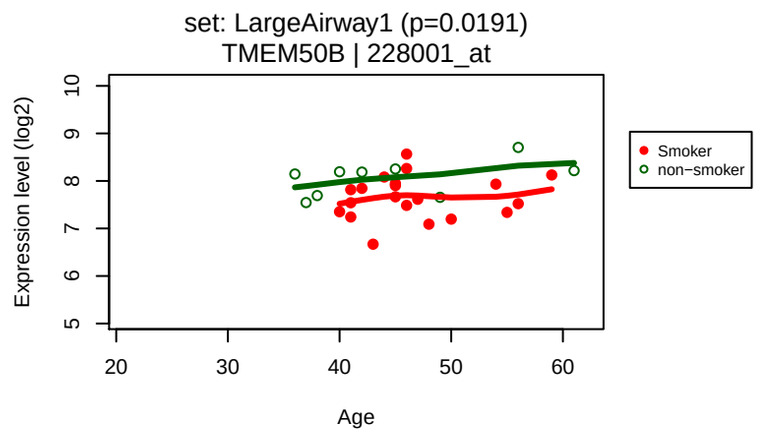

Gene expression of smokers and nonsmokers
gene: TMEM50B | 228001_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.3414
p=0.6159
p=0.9122
p=0.0191
p=0.1738
p=0.3166
p=0.1542
p=0.0233
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

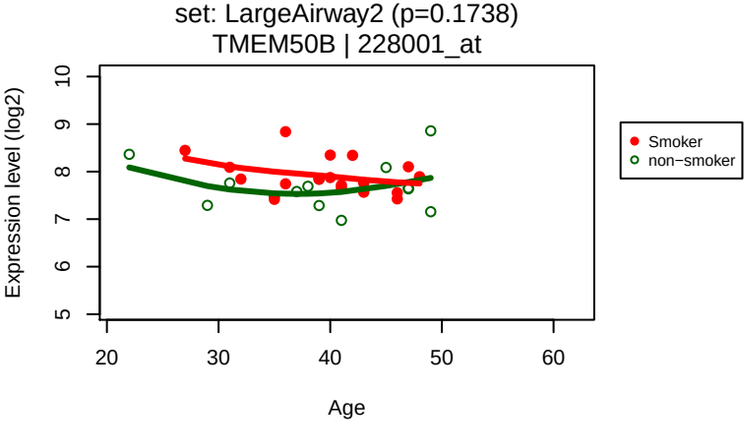

Gene expression ratio (smokers/nonsmokers)
gene: TMEM50B | 228001_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3414
p=0.6159
p=0.9122
p=0.0191
p=0.1738
p=0.3166
p=0.1542
p=0.0233
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

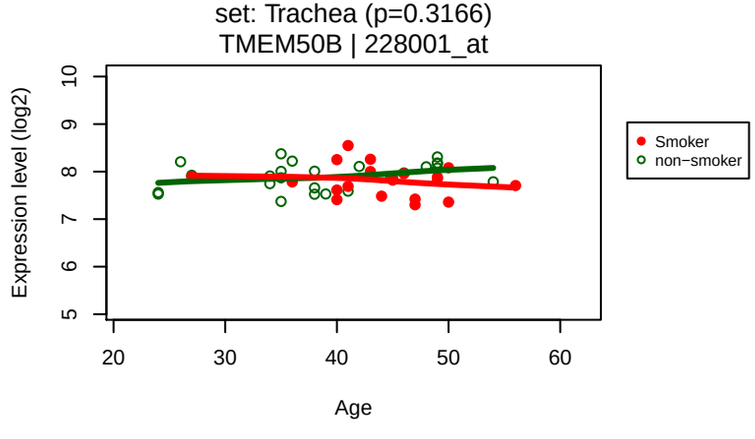

Gene expression ratio in Males
gene: TMEM50B | 228001_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6433
p=0.469
p=0.2634
p=0.0725
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: TMEM50B | 228001_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8169
p=0.1672
p=0.8982
p=0.2322
SmallAirway2
LargeAirway2
Trachea
Macrophage2

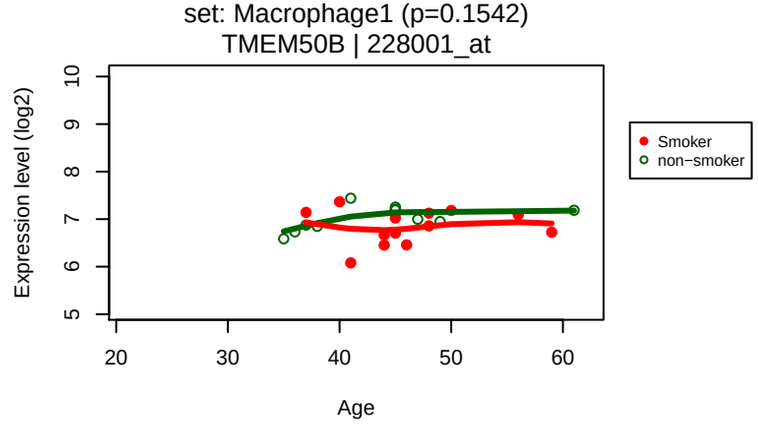

set: SmallAirway1 (p=0.3414)
TMEM50B | 228001_at
Expression level (log2)
Age
Smoker
non−smoker

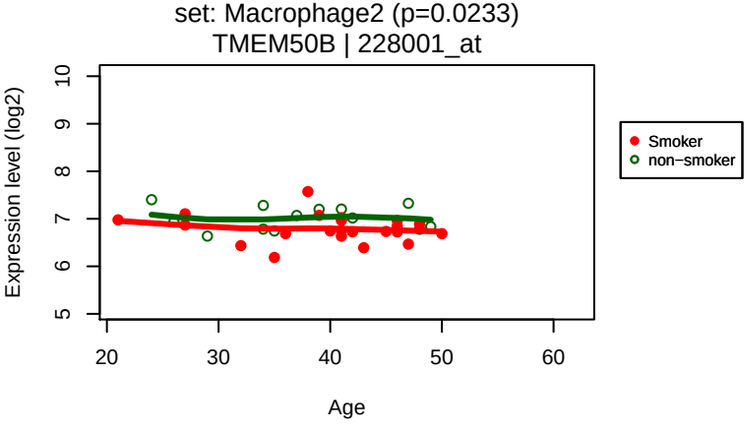

set: SmallAirway2 (p=0.6159)
TMEM50B | 228001_at
Expression level (log2)
Age
Smoker
non−smoker

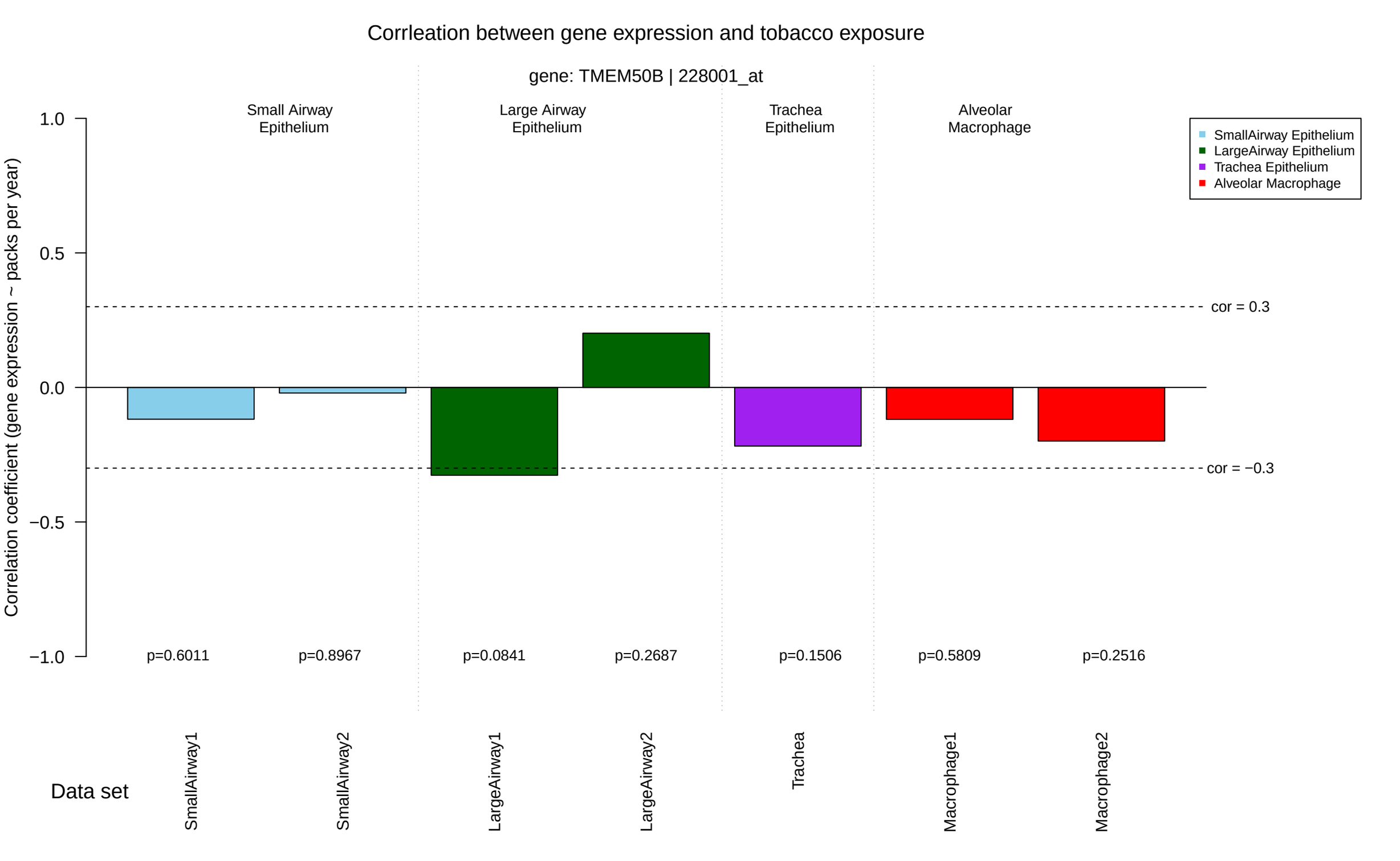

set: LargeAirway1 (p=0.0191)
TMEM50B | 228001_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.1738)
TMEM50B | 228001_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.3166)
TMEM50B | 228001_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.1542)
TMEM50B | 228001_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0233)
TMEM50B | 228001_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: TMEM50B | 228001_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.6011
p=0.8967
p=0.0841
p=0.2687
p=0.1506
p=0.5809
p=0.2516
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2